

A physics-informed matrix engine bypasses scaling relations in alkyne-poisoned catalytic interfaces — Part 1

Aniema Ekemini*

Department of Science, Southern British High School, Calabar, CRS, Nigeria

Ededet Asuquo

Department of Science, Southern British High School, Calabar, CRS, Nigeria

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Abstract

High-throughput catalyst design is frequently constrained by linear scaling relations, which enforce a strict compromise between catalytic activity and interface deactivation. In the selective hydrogenation of olefins, terminal alkyne impurities cause severe active-site poisoning via irreversible σ -acetylide formation, yet traditional static electronic descriptors fail to distinguish this destructive path from benign π -coordination. Here, we report a physical tight-binding framework based on a modified three-center Hückel-Coulson Hamiltonian that resolves this limitation by evaluating quantum mechanical conservation principles in sub-millisecond cycles. By deriving dynamic wavefunction-based descriptors— π -backbonding susceptibility (ΔE_π) and Coulson charge redistribution (Δq_{C1})—our matrix engine successfully isolates and maps the electronic phase boundaries governing poisoning pathways. Screening a digital library of 300,000 distinct catalyst-reactant configurations uncovers a class of coinage-metal single-atom alloys and spatially confined single-atom catalysts that completely bypass classical scaling trends through local charge inversion. Kinetic Monte Carlo simulations driven by these physics-embedded metrics predict near-infinite active-site preservation under realistic process environments. This computationally scalable framework replaces black-box correlations with explicit chemical intuition, offering an adaptable paradigm for navigating complex catalytic phase spaces.

1 Introduction

High-throughput computational screening has emerged as a cornerstone of modern catalyst discovery, yet its velocity is fundamentally throttled by the persistent challenge of catalyst deactivation [1]. In industrial chemical purification streams—most notably the selective hydrogenation of ethylene feeds—the presence of trace terminal alkynes (such as acetylene or propyne) presents a severe poisoning risk to transition metal active sites [2]. These impurities undergo rapid, competitive adsorption, triggering irreversible active-site blocking that degrades catalyst lifetime and product selectivity [3].

Historically, navigating this design space has relied on linear scaling relations, which assert a strict thermodynamic correlation between the binding energies of reaction intermediates and those of poisonous side-products [4]. Consequently, efforts to optimize the catalytic interface for

*Corresponding author. Email: ekemjah5@gmail.com

alkene production inevitably exacerbate its vulnerability to permanent alkyne deactivation. This compromise reveals a fundamental limitation in our current descriptor paradigms. Traditional static metrics, such as the surface d -band center (ε_d), assume an unyielding electronic state during the coordination process. In doing so, they fail to capture the dynamic orbital hybridization, local charge redistribution, and symmetry-allowed mixing that occur as a reactant approaches an active site.

To bypass the limitations of linear scaling relations, recent computational discovery pipelines have shifted toward high-throughput screening using density functional theory (DFT). While DFT offers detailed insight into electronic structures, its high computational demand makes it impractical for exploring vast, multi-component chemical spaces or complex alloy surfaces [5]. This issue has led to the widespread adoption of data-driven machine learning (ML) models [6].

However, pure ML screening approaches often act as uninterpretable black boxes, mapping structural inputs directly to adsorption energies without providing physical or chemical context [7]. Many of these models lack explicit physical constraints, such as quantum mechanical conservation principles or orbital symmetry rules [8]. When applied to complex tasks like alkyne coordination, generic ML models frequently fail to distinguish between non-destructive, reversible π -coordination and destructive, irreversible σ -acetylide formation. This systematic blind spot highlights the urgent need for a screening method that combines the speed of data-driven models with the physical rigor and interpretability of quantum mechanics.

References

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Data and Code Availability

The high-throughput screening dataset containing the calculated quantum metrics (ΔE_π and Δq_{C1}) for the 300,000 catalyst–reactant configurations, along with the foundational periodic DFT parameters, has been permanently archived on Zenodo under DOI: 10.5281/zenodo.21196337. The complete, self-contained Python script used to verify these analytical metrics and drive the screening matrix engine is provided below: